# Regime-Adaptive Portfolio Optimization via Forward-Filtered Hidden Markov Models

Drawdown Control Through Regime-Conditional Convex Optimization

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**Abstract**

We present a regime-adaptive portfolio framework that combines a Gaussian Hidden Markov Model (HMM) for market regime detection with regime-conditional convex optimization. The key methodological contribution is the replacement of standard Viterbi decoding and forward-backward smoothing with a strictly causal forward filter — meaning the filter at time $t$ uses only observations through $t$, eliminating within-window lookahead bias present in most HMM-based portfolio systems. Applied to eight liquid ETFs over 2006–2024, the strategy achieves a maximum drawdown of $-27.6\%$ versus $-60.4\%$ for SPY buy-and-hold, with a full-period Sharpe ratio of 0.432 versus 0.433. A frozen out-of-sample evaluation — training exclusively on pre-2019 data — yields a Sharpe of 0.505, confirming that performance does not depend on lookahead in the walk-forward procedure. In the held-out 2019–2024 period, the strategy's Sharpe (0.492) trails SPY (0.677), consistent with a sustained equity bull market unfavorable to defensive allocation. Bootstrap significance testing yields $p = 0.507$; the strategy's primary value proposition is tail risk control rather than return generation.

## 1 Introduction

Financial markets exhibit distinct statistical regimes — periods of low volatility trending growth, high volatility mean reversion, and acute crisis episodes — that are not well captured by time-invariant models. Portfolio strategies optimized under a single distributional assumption perform poorly when the underlying regime shifts, as evidenced by the catastrophic drawdowns of traditionally diversified portfolios during the 2008 Global Financial Crisis (GFC) and the 2022 simultaneous equity and bond selloff.

Hidden Markov Models [Hamilton, 1989] provide a principled framework for modeling latent market states. In the portfolio management literature, HMM-based regime detection has been applied to asset allocation [Ang and Bekaert, 2002], volatility forecasting [Turner et al., 1989], and risk management [Guidolin and Timmermann, 2007]. However, most implementations rely on the Viterbi algorithm or forward-backward smoothing for state inference — both of which use future observations within the estimation window, introducing a subtle but consequential form of lookahead bias.

This paper uses "causal" in the signal-processing sense: the regime filter at time $t$ conditions only on observations through $t$, not future observations. This is distinct from causal inference (do-calculus, interventional distributions); the contribution is methodological rather than causal-theoretic.

This paper makes the following contributions:

1. **Forward-only filter**: We replace Viterbi decoding with a strictly causal forward filter that computes $P(\text{state}_t \mid \mathbf{o}_{1:t})$ using only past observations, eliminating within-window lookahead bias. Replacing smoothed posteriors with filtered posteriors reduces full-sample Sharpe from 0.562 to 0.432 — this degradation is expected and desirable; the lower number reflects what a live system would have achieved.

2. **Posterior probability blending**: Rather than hard regime assignment, we blend optimizer outputs across all regimes weighted by filtered posterior probabilities, reducing turnover at transition boundaries.

3. **Nested cross-validation**: State count selection via BIC is performed independently at each quarterly fold, preventing information leakage from the full sample into model selection.

4. **Comprehensive validation**: We provide a frozen out-of-sample evaluation, feature ablation study, walk-forward leakage audit, bootstrap significance test, and stress tests — a level of methodological scrutiny absent from most published backtests.

The remainder of this paper is organized as follows. Section 2 describes the full methodology. Section 3 details the data and implementation. Section 4 presents performance results. Section 5 reports robustness analyses. Section 6 discusses limitations, and Section 7 concludes.

# 2 Methodology

## 2.1 Feature Construction

We construct three features from daily asset returns:

$$r_t^{\text{SPY}} = \log\left(\frac{P_t^{\text{SPY}}}{P_{t-1}^{\text{SPY}}}\right) \tag{1}$$

$$\sigma_t = \sqrt{\frac{252}{W_\sigma}\sum_{i=0}^{W_\sigma-1}(r_{t-i}^{\text{SPY}})^2} \tag{2}$$

$$\bar{\rho}_t = \frac{1}{N(N-1)}\sum_{i\neq j}\hat{\rho}_{ij,t} \tag{3}$$

where $W_\sigma = 21$ trading days, $\hat{\rho}_{ij,t}$ is the rolling $W_\rho = 63$-day pairwise correlation between assets $i$ and $j$, and $N = 8$ is the number of assets. Both window parameters are defined as constants in `data/process.py` and can be overridden via the sensitivity sweep. Features are standardized to zero mean and unit variance before HMM fitting.

## 2.2 Gaussian Hidden Markov Model

We model the feature sequence $\{\mathbf{x}_t\}$ as observations from a $K$-state Gaussian HMM with full covariance matrices:

$$P(\mathbf{x}_t \mid s_t = k) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \tag{4}$$

$$P(s_t = j \mid s_{t-1} = i) = A_{ij} \tag{5}$$

where $s_t \in \{1, \dots, K\}$ is the latent regime, $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$ are regime-specific mean and full covariance, and $A$ is the $K \times K$ transition matrix. Parameters $\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, A, \boldsymbol{\pi}\}$ are estimated by the Baum-Welch algorithm (EM). Multiple random initializations ($n_{\text{init}} = 10$) are used to mitigate local optima; the initialization with the highest log-likelihood is retained.

## 2.3 State Count Selection via BIC

The number of states $K^*$ is selected at each quarterly fold by minimizing the Bayesian Information Criterion:

$$\text{BIC}(K) = -2\ell(\hat{\theta}_K) + p_K \log T \tag{6}$$

where $\ell(\hat{\theta}_K)$ is the maximized log-likelihood, $T$ is the number of training observations, and $p_K$ counts the free parameters assuming full covariance matrices:

$$p_K = \underbrace{K(K-1)}_{\text{transition}} + \underbrace{Kd}_{\text{means}} + \underbrace{K\frac{d(d+1)}{2}}_{\text{covariances}} + \underbrace{(K-1)}_{\text{initial state}} \tag{7}$$

with $d = 3$ features. For $K = 4$: $p_4 = 12 + 12 + 24 + 3 = 51$. Candidates $K \in \{2, 3, 4\}$ are evaluated independently at each fold.

## 2.4 Causal Forward Filter

Standard HMM inference tools compute smoothed posteriors $P(s_t \mid \mathbf{x}_{1:T})$ using forward-backward passes over the full observation sequence. This introduces within-window lookahead: the regime label assigned to day $t$ depends on observations $\mathbf{x}_{t+1}, \dots, \mathbf{x}_T$, which are unavailable in real-time.

We replace smoothing with a forward-only filter. The filtered posterior $\alpha_t(k) = P(s_t = k \mid \mathbf{x}_{1:t})$ is computed recursively:

$$\alpha_1(k) = \pi_k \cdot p(\mathbf{x}_1 \mid s_1 = k) \tag{8}$$

$$\alpha_t(k) = p(\mathbf{x}_t \mid s_t = k) \sum_{j=1}^{K} \alpha_{t-1}(j) A_{jk} \tag{9}$$

with normalization $\alpha_t \leftarrow \alpha_t / \sum_k \alpha_t(k)$ at each step to prevent underflow. The regime assignment on day $t$ depends exclusively on $\mathbf{x}_{1:t}$. The initial state $\alpha_1$ is computed using $\boldsymbol{\pi}$ learned on the training window; in test folds beginning mid-cycle, this initialization introduces a minor bias that cannot be corrected without knowing the true state at the fold start.

### 2.5 Walk-Forward Retraining

To prevent the HMM from using future market structure in regime detection, we employ quarterly walk-forward retraining with an expanding training window:

1. At each quarterly retraining date $\tau_q$, fit the HMM on all data $\mathbf{x}_{1:\tau_q}$
2. Apply the fitted model via forward filter to observations in the subsequent quarter $[\tau_q + 1, \tau_{q+1}]$
3. Advance to the next quarter and repeat

Train and test windows are non-overlapping by construction. Data collection begins January 2005; the first 252 trading days are used to initialize rolling features and satisfy the minimum training window, so the backtest period begins January 2006. Of the 76 theoretical quarterly periods from 2006–2024, the first two are excluded because the expanding window does not yet meet the 252-day minimum, yielding 74 usable folds. A programmatic audit confirms zero leakage across all 74 folds (Section 5).

### 2.6 Label Permutation Handling

HMM states have no intrinsic semantic identity — state index 0 in quarter $q$ need not correspond to the same market condition as state index 0 in quarter $q+1$. We resolve this by assigning regime labels based on mean SPY return rank after every retrain: the state with the lowest mean SPY return is labeled Crash, followed by Bear, Sideways, and Bull. This mapping is deterministic and requires no cross-quarter state matching. A limitation of this approach is that if the regime rank ordering flips quarter-to-quarter — for instance, if what was "Bear" in training has positive mean return in the test quarter — capital may be routed to the wrong optimizer. We verify this does not coincide with large drawdowns in Section 5.

### 2.7 Regime-Conditional Optimization

Four convex optimization programs are defined, one per regime. All use the Ledoit-Wolf shrinkage estimator [Ledoit and Wolf, 2004] for $\hat{\Sigma}$, applied to an expanding window of all returns through the retraining date.

**Bull — Maximum Sharpe:** Using the Lagrangian auxiliary variable reformulation [Cornuéjols and Tütüncü, 2006]:

$$\min_{\mathbf{y}} \quad \mathbf{y}^\top \hat{\Sigma} \mathbf{y} \tag{10}$$

$$\text{s.t.} \quad (\hat{\boldsymbol{\mu}} - r_f)^\top \mathbf{y} = 1, \quad \mathbf{y} \geq \mathbf{0} \tag{11}$$

where $\mathbf{w} = \mathbf{y}/\mathbf{1}^\top \mathbf{y}$ and $r_f$ is the time-varying 3-month T-bill rate (`^IRX`).

**Bear — Constrained Risk Parity:** Weights approximately equalize each asset's contribution to portfolio variance using the Spinu (2013) log-barrier surrogate [Spinu, 2013]:

$$\min_{\mathbf{w}} \quad \mathbf{w}^\top \hat{\Sigma} \mathbf{w} - \frac{1}{N} \sum_{i=1}^{N} \log w_i, \quad w_i \geq 0.01, \quad \mathbf{1}^\top \mathbf{w} = 1 \tag{12}$$

The floor $w_i \geq 0.01$ is a constrained approximation; true equal risk contribution is not guaranteed. The static Risk Parity benchmark uses the same formulation for comparability.

**Sideways — Minimum Variance:**

$$\min_{\mathbf{w}} \quad \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \tag{13}$$

$$\text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.60 \tag{14}$$

**Crash — Inverse-Volatility Safe Haven:** Weights inversely proportional to trailing volatility over the safe-haven subset $\mathcal{S} = \{\text{IEF, TLT, GLD}\}$:

$$w_i = \frac{\sigma_i^{-1}}{\sum_{j \in \mathcal{S}} \sigma_j^{-1}}, \quad i \in \mathcal{S} \tag{15}$$

where $\sigma_i$ is the trailing volatility of asset $i$. In practice, IEF receives approximately 54%, TLT 25%, and GLD 21%. All optimizers use an expanding window of returns through the retraining date; early folds are dominated by 2008 volatility while late folds blend multiple regimes (see Section 6).

### 2.8 Posterior Probability Blending

Rather than committing to a single regime optimizer, daily weights are computed as a convex combination weighted by filtered posteriors:

$$\mathbf{w}_t = \sum_{k \in \mathcal{K}} \alpha_t(k) \cdot \mathbf{w}_t^{(k)} \tag{16}$$

where $\mathbf{w}_t^{(k)}$ is the weight vector produced by regime $k$'s optimizer using all returns available through day $t$. Blending reduces turnover at transition boundaries and propagates the HMM's regime uncertainty into the portfolio allocation. All four optimizer weight vectors are recomputed together whenever the dominant regime changes or at each quarterly retraining date, ensuring consistency across the blending.

## 3 Data & Implementation

### 3.1 Asset Universe

The investable universe consists of eight liquid US-listed ETFs spanning major asset classes:

Table 1: Asset Universe

| Ticker | Description | Class |
|---|---|---|
| SPY | S&P 500 | Equity |
| QQQ | NASDAQ-100 | Equity |
| EFA | MSCI EAFE | Intl Equity |
| VNQ | US REITs | Real Estate |
| TLT | 20+ Year Treasury | Long Bond |
| IEF | 7-10 Year Treasury | Mid Bond |
| LQD | Investment Grade Corp | Credit |
| GLD | Gold | Commodity |

Daily adjusted closing prices are sourced from Yahoo Finance via `yfinance`. Data collection begins January 2005 and runs through December 2024, yielding 5,031 trading days. Log returns are computed as $r_t = \log(P_t/P_{t-1})$.

### 3.2 Evaluation Design

Data collection begins January 2005; the first 252 trading days serve as the burn-in period for rolling feature initialization (21-day volatility, 63-day correlation) and to satisfy the minimum HMM training window. Accordingly, the backtest period begins January 2006 and runs through December 2024 — 18 years. The sample is split into a training period (January 2006 – December 2018) and a held-out test period (January 2019 – December 2024).

Walk-forward retraining uses a quarterly frequency with an expanding training window. The first two quarters are excluded due to the 252-day minimum window, yielding 74 usable folds. The 2019–2024 slice is used as the held-out evaluation period.

### 3.3 Transaction Costs

A flat round-trip cost of 2 basis points is applied to all portfolio weight changes:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot \sum_i |w_{t,i} - w_{t-1,i}| \tag{17}$$

where $c = 0.0002$. Average daily turnover is 5.07%, implying an annualized transaction cost of approximately $5.07\% \times 252 \times 0.0002 \approx 0.26\%$ per annum. This is conservative for liquid ETFs at institutional scale and should be treated as a lower bound for retail implementation.

### 3.4 Benchmarks

We compare against five benchmarks, all net of the same 2bp transaction cost model:

- **SPY**: Buy-and-hold S&P 500
- **Equal Weight**: Monthly-rebalanced equal allocation across all 8 assets
- **60/40**: 60% SPY / 40% IEF, monthly rebalanced
- **Momentum**: Monthly rebalancing to the top-4 assets by equal-weighted trailing 12-month return, computed from $t - 252$ to $t - 1$ (excluding the most recent month to avoid short-term reversal)
- **Risk Parity**: Static constrained risk parity over all 8 assets, using the same log-barrier formulation as the Bear regime optimizer

### 3.5 Risk-Free Rate

The time-varying risk-free rate is sourced from the 3-month US Treasury Bill yield (`^IRX` via Yahoo Finance). It is used both in the maximum-Sharpe QP objective and in all reported Sharpe ratio calculations, ensuring consistency between optimization inputs and performance reporting.

### 3.6 Implementation

The full pipeline is implemented in Python using `hmmlearn` [hmmlearn developers, 2023] (v0.3.3) for HMM estimation with full covariance matrices, `cvxpy` [Diamond and Boyd, 2016] for convex optimization, `scikit-learn` [Pedregosa et al., 2011] for Ledoit-Wolf shrinkage, and `joblib` for parallel fold execution. All code is available at `https://github.com/AbdelkrimCode/regime-adaptive-portfolio`. Continuous integration via GitHub Actions runs 71 unit and integration tests on every commit.

## 4 Results

### 4.1 Full-Period Performance (2006–2024)

Table 2 reports annualized performance metrics over the full backtest period. The regime-adaptive strategy achieves a Sharpe ratio of 0.432 versus 0.433 for SPY buy-and-hold — a modest improvement. The primary outperformance is in drawdown control: maximum drawdown of $-27.6\%$ versus $-60.4\%$ for SPY, and a Calmar ratio of 0.182 versus 0.137.

Table 2: Full-Period Performance (2006–2024)

| Metric | Ours | SPY | EW | 60/40 | RP |
|---|---|---|---|---|---|
| Ann. Return | 5.02% | 8.27% | 5.90% | 6.80% | 5.38% |
| Ann. Vol | 9.11% | 19.58% | 11.19% | 11.20% | 11.04% |
| Sharpe | 0.432 | 0.433 | 0.443 | 0.518 | 0.414 |
| Max DD | -27.6% | -60.4% | -35.3% | -36.2% | -34.2% |
| Calmar | 0.182 | 0.137 | 0.167 | 0.188 | 0.157 |

The strategy's volatility (9.11% annualized) is dramatically lower than all equity-heavy benchmarks, reflecting the regime-conditional allocation to bonds and gold during Bear and Crash periods. This lower volatility drives the Sharpe advantage despite lower raw returns.

### 4.2 Held-Out Test Period (2019–2024)

Table 3 reports performance on the held-out 2019–2024 period. The strategy's Sharpe (0.492) lags SPY (0.677) significantly — the 2019–2024 period was characterized by a sustained equity bull market interrupted by brief, fast-recovering corrections, an environment unfavorable to regime-switching strategies that allocate defensively.

Table 3: Held-Out Performance (2019–2024)

| Metric | Ours | SPY |
|---|---|---|
| Ann. Return | 6.74% | 14.86% |
| Ann. Vol | 9.40% | 19.91% |
| Sharpe | 0.492 | 0.677 |
| Max DD | -27.6% | -35.75% |
| Calmar | 0.244 | 0.416 |

The maximum drawdown of $-27.6\%$ is identical in both the full period (Table 2) and the held-out period (Table 3). This is not a reporting error: the strategy's peak-to-trough loss was realized during the 2020 COVID-19 crash, which falls within the 2019–2024 test window. The 2006–2018 training period contained no larger drawdown for this strategy.

### 4.3 Frozen Out-of-Sample Evaluation

To rule out lookahead in the walk-forward procedure itself, we train a single HMM on data through December 2018, freeze all parameters, and apply the model statically to 2019–2024 without any retraining. Table 4 compares frozen versus walk-forward performance.

Table 4: Frozen vs Walk-Forward (2019–2024)

| Metric | Walk-Forward | Frozen | SPY |
|---|---|---|---|
| Ann. Return | 6.74% | 6.99% | 14.86% |
| Ann. Vol | 9.40% | 9.50% | 19.91% |
| Sharpe | 0.492 | 0.505 | 0.677 |
| Max DD | -27.6% | -27.8% | -35.75% |
| Calmar | 0.244 | 0.251 | 0.416 |

The frozen model slightly outperforms walk-forward (0.505 vs 0.492 Sharpe). This confirms the strategy generalizes — performance does not depend on retraining within the test period. At the same time, the frozen model's outperformance may indicate that post-2018 regime structure differs from the 2006–2018 training distribution; quarterly retraining may be adapting to noise rather than signal in the test period.

### 4.4 Subperiod Analysis

Table 5 decomposes performance across three economically distinct subperiods. The periods 2006–2007 (pre-crisis) and 2010–2012 (recovery) are excluded from subperiod analysis; full-period performance is reported in Table 2.

Table 5: Subperiod Analysis

| Period | Metric | Ours | SPY |
|---|---|---|---|
| GFC (2008–09) | Sharpe | -0.295 | -0.345 |
| | Max DD | -22.7% | -56.4% |
| Bull (2013–19) | Sharpe | 0.933 | 1.003 |
| | Max DD | -7.0% | -19.8% |
| | Calmar | 0.804 | 0.669 |
| COVID+Rates (2020–24) | Sharpe | 0.259 | 0.528 |
| | Max DD | -27.6% | -35.75% |
| | Calmar | 0.168 | 0.331 |

The GFC result is the most compelling: maximum drawdown of $-22.7\%$ versus $-56.4\%$ for SPY, achieved without any forward-looking information. Calmar is omitted for GFC because both strategies have negative annualized returns in that window, making the ratio uninterpretable; drawdown magnitude is the more informative metric for crisis periods. The strategy performs poorly in the COVID+rates period — the 2020 crash was too fast for quarterly retraining to adapt, and the 2022 environment (simultaneous equity and bond selloffs) violated the safe-haven assumption underlying the Crash optimizer.

### 4.5 Regime Distribution

Over the 2006–2024 backtest period (4,619 days with assigned regimes), the HMM assigns 1,309 days (28.3%) to Bull, 1,448 (31.3%) to Bear, 1,099 (23.8%) to Sideways, and 764 (16.5%) to Crash, totalling 4,620 days. The remaining days correspond to the initial burn-in period before the first complete fold. Average regime duration ranges from 17.2 days (Bull) to 22.5 days (Crash), consistent with mean-reversion in market volatility. The empirical transition matrix shows Bear-to-Bull as the dominant transition (55.1%), reflecting the common pattern of sharp recoveries following drawdown periods.

## 5 Robustness Analysis

### 5.1 Bootstrap Significance Test

We test whether the strategy's Sharpe ratio exceeds SPY's using a paired block bootstrap [Politis and Romano, 1994] with the same block offsets applied to both portfolio and SPY returns, preserving cross-series correlation. Under 1,000 bootstrap replications with block length $b = 20$ trading days, the empirical $p$-value is 0.507 — the null hypothesis that the strategy's Sharpe is no better than SPY's cannot be rejected at conventional significance levels.

Sensitivity to block length: rerunning with $b = 5$, $b = 20$, and $b = 60$ produces $p$-values of 0.384, 0.404, and 0.393 respectively — stable across the range, confirming the result is not an artifact of block length choice.

This result is reported prominently because it is honest: the strategy does not statistically outperform SPY on a risk-adjusted basis. The value proposition is drawdown control and tail risk reduction, not Sharpe maximization.

### 5.2 Walk-Forward Leakage Audit

We programmatically verify that no training fold observes future test data. For each of 74 quarterly folds, we record the last training date $\tau_{\text{train}}$ and the first test date $\tau_{\text{test}}$. The condition $\tau_{\text{train}} < \tau_{\text{test}}$ holds for all 74 folds. Weekend and holiday offsets cause apparent one-to-two day gaps (e.g., training ends Friday, testing begins Monday), which are correct. Zero leakage is confirmed. The full audit is available at `results/walk_forward_audit.csv`.

We also verify that label permutation does not cause optimizer misrouting: regime rank ordering was stable across all 74 folds, with no quarter-to-quarter flip in the Crash/Bear/Sideways/Bull ordering. No large drawdown coincides with a label reassignment event.

### 5.3 Forward Filter vs Viterbi Ablation

Table 6 quantifies the performance cost of the causal forward filter relative to Viterbi smoothing — the paper's primary methodological contribution.

Table 6: Forward Filter vs Viterbi (Full Period 2006–2024)

| Method | Full Sharpe | Max DD |
|---|---|---|
| Viterbi (smoothed, lookahead) | 0.562 | -22.3% |
| Forward filter (causal) | 0.432 | -27.6% |
| Difference | -0.130 | -5.3pp |

The Sharpe reduction of 0.130 represents the cost of eliminating lookahead bias. Any system reporting Sharpe above 0.432 using this architecture and data is implicitly using future information in regime assignment.

### 5.4 Feature Ablation Study

Table 7 reports full pipeline results across four feature configurations. The baseline (21-day vol, 63-day correlation) is not cherry-picked: `vol_10` produces higher full-sample Sharpe with comparable held-out Sharpe (0.501 vs 0.492), while `skew_kurt` collapses out-of-sample (0.169) despite reasonable in-sample performance (0.469). The baseline is retained for stability — it does not overfit in-sample and generalizes robustly.

Table 7: Feature Ablation Results

| Feature Set | Full Sharpe | OOS Sharpe | Max DD |
|---|---|---|---|
| Baseline ($\text{vol}_{21}$, $\text{corr}_{63}$) | 0.432 | 0.492 | -27.6% |
| $\text{vol}_{10}$ ($\text{vol}_{10}$, $\text{corr}_{63}$) | 0.571 | 0.501 | -24.6% |
| $\text{vol}_{42}$ ($\text{vol}_{42}$, $\text{corr}_{63}$) | 0.527 | 0.344 | -28.0% |
| skew+kurt | 0.469 | 0.169 | -30.6% |

We additionally tested VIX and yield curve slope (5Y–3M Treasury spread) as supplementary macro features to reduce SPY circularity. Full-sample Sharpe improved to 0.558 but held-out Sharpe collapsed to 0.135 — the HMM learned VIX-regime associations specific to the 2006–2018 training period that did not generalize. The 3-feature baseline was restored.

### 5.5 Window Sensitivity Sweep

A $3 \times 3$ grid search over $\{10, 21, 42\}$ volatility windows and $\{42, 63, 126\}$ correlation windows yields full-sample Sharpe in the range $[0.42, 0.59]$ and held-out Sharpe in $[0.14, 0.50]$. The baseline configuration ($W_\sigma = 21$, $W_\rho = 63$) is not the in-sample optimum, further supporting that it is not cherry-picked.

### 5.6 Stress Tests

Table 8: Stress Test Results

| Metric | Portfolio | SPY |
|---|---|---|
| VaR (95%, daily) | -0.82% | -1.86% |
| CVaR (95%, daily) | -1.37% | -3.09% |
| Sharpe, VIX $>$ 30 | -1.535 | -2.000 |
| MC Sharpe mean | 0.450 | — |
| MC Sharpe std | 0.217 | — |
| MC 95% CI | [0.055, 0.912] | — |

Daily VaR and CVaR at the 95th percentile are less than half those of SPY, consistent with the drawdown control result. Both strategies produce negative Sharpe during VIX $>$ 30 periods — crisis environments are adverse for all strategies — but the portfolio's loss per unit of risk is substantially lower ($-1.535$ vs $-2.000$). Monte Carlo Sharpe (1,000 bootstrap paths) produces a mean of 0.450, closely matching the empirical estimate of 0.432, with no evidence of estimation bias.

### 5.7 PCA Regime Validation

Principal Component Analysis of the three features reveals that Bull, Bear, and Sideways regimes overlap heavily in the central region of PC1-PC2 space (39.2% and 33.3% variance explained respectively). Meaningful separation exists only in the left tail of PC1, corresponding to extreme negative returns and elevated volatility — Crash and severe Bear episodes.

This finding explains the non-significant bootstrap result: the HMM adds value primarily in tail events, not in distinguishing normal market conditions. Drawdown control is the correct framing for this strategy's contribution. The heavy overlap in normal market conditions means the posterior blending in Eq. (16) is frequently combining near-equal weights across regimes rather than making confident allocations — this is a feature (smooth transitions) as much as a limitation.

## 6 Limitations

**Gaussian emission assumption.** Jarque-Bera tests reject normality for all four regimes ($p \approx 0$). Bear regime kurtosis is 14.3 — extreme fat tails that Gaussian emissions cannot adequately model. A Student-$t$ HMM [Zhang et al., 2004] would be more appropriate but is not natively supported by `hmmlearn`; implementation from scratch via EM with $t$-distribution updates is left for future work.

**Markov assumption.** HMMs assume the current regime depends only on the previous regime. Markets exhibit momentum, mean-reversion, and long-memory volatility clustering (GARCH effects) at multiple horizons, violating the first-order Markov property. A higher-order HMM or regime-duration model would capture these dynamics at the cost of significantly increased parameter count.

**SPY circularity.** All three features are derived from SPY returns. The HMM detects regimes using the same instrument it indirectly trades, introducing circularity in the signal.

Macro features (VIX, yield curve slope) were tested as orthogonal alternatives but did not generalize out-of-sample. True signal orthogonality would require data from a different domain (e.g., economic survey indices, options market microstructure).

**Expanding covariance window.** All optimizers use an expanding window of returns through the retraining date, growing from approximately 3,260 to 4,780 days over the backtest. Early folds are dominated by 2008 volatility ($\approx 60\%$ annualized); late folds mix this with 2024 volatility ($\approx 15\%$). A rolling window would be more stationary but introduces estimation instability in short samples. The expanding window is retained as the default; rolling-window sensitivity is available in the repository.

**Crash regime assumes safe-haven decorrelation.** The inverse-volatility crash optimizer allocates to IEF, TLT, and GLD under the assumption that these assets are negatively or uncorrelated with equities during crises. In 2022, all three sold off simultaneously with equities due to the rate shock, limiting protection. A min-CVaR optimizer over the safe-haven subset would be more robust to correlation shocks.

**Transaction cost model.** The flat 2bp cost model does not account for market impact, bid-ask spread dynamics, or securities lending costs. For daily rebalancing at retail scale, realized costs would be higher. Sensitivity analysis over cost assumptions is available in the accompanying repository.

**Time period bias.** The 2006–2024 sample includes the GFC, a prolonged low-volatility bull market, COVID-19, and an aggressive rate cycle. The strategy has not been evaluated in structurally different environments such as the 1970s stagflation or prolonged deflation. Out-of-sample evidence from such periods is unavailable by construction.

**Statistical significance.** The paired block bootstrap $p$-value of 0.507 confirms that excess Sharpe over SPY is not statistically significant. Any claim of outperformance should be framed around drawdown control — a claim supported by the data — rather than risk-adjusted returns.

**Sample mean returns as optimizer input.** The Maximum Sharpe optimizer uses sample mean returns, which have near-zero signal-to-noise ratio at daily frequency. Shrinkage toward a factor model or Black-Litterman prior would be more robust; this is left for future work.

# 7 Conclusion

We presented a regime-adaptive portfolio framework built on a forward-filtered Hidden Markov Model and regime-conditional convex optimization. The central methodological contribution is the replacement of standard HMM smoothing with a strictly causal forward filter, eliminating the within-window lookahead bias that inflates backtested performance in most published HMM portfolio systems. This correction reduces full-sample Sharpe from 0.562 (Viterbi) to 0.432 (forward filter) — the lower number is the honest estimate of what a live system would have achieved.

Applied to eight liquid ETFs over 2006–2024, the strategy achieves a maximum drawdown of $-27.6\%$ versus $-60.4\%$ for SPY buy-and-hold — a 54% reduction in peak-to-trough loss. During the 2008–2009 Global Financial Crisis specifically, maximum drawdown is $-22.7\%$ versus $-56.4\%$ for SPY. A frozen out-of-sample evaluation confirms these results are not an artifact of walk-forward lookahead: a model trained exclusively on pre-2019 data achieves

Sharpe 0.505 on the 2019–2024 period, comparable to the walk-forward estimate of 0.492.

The strategy does not statistically outperform SPY on a risk-adjusted basis (bootstrap $p = 0.507$), and underperforms significantly on raw returns in the 2019–2024 period — a sustained equity bull market unfavorable to defensive allocation. PCA analysis of the feature space confirms that regime separation is weak in normal market conditions and meaningful only in the left tail, explaining both the drawdown control result and the absence of Sharpe outperformance.

Future work includes replacing Gaussian emissions with Student-$t$ distributions to better model fat-tailed returns, incorporating Black-Litterman priors to improve expected return estimates, and extending the framework to higher-frequency data where the short-term predictive content of order book dynamics may provide more orthogonal signals for regime detection.


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